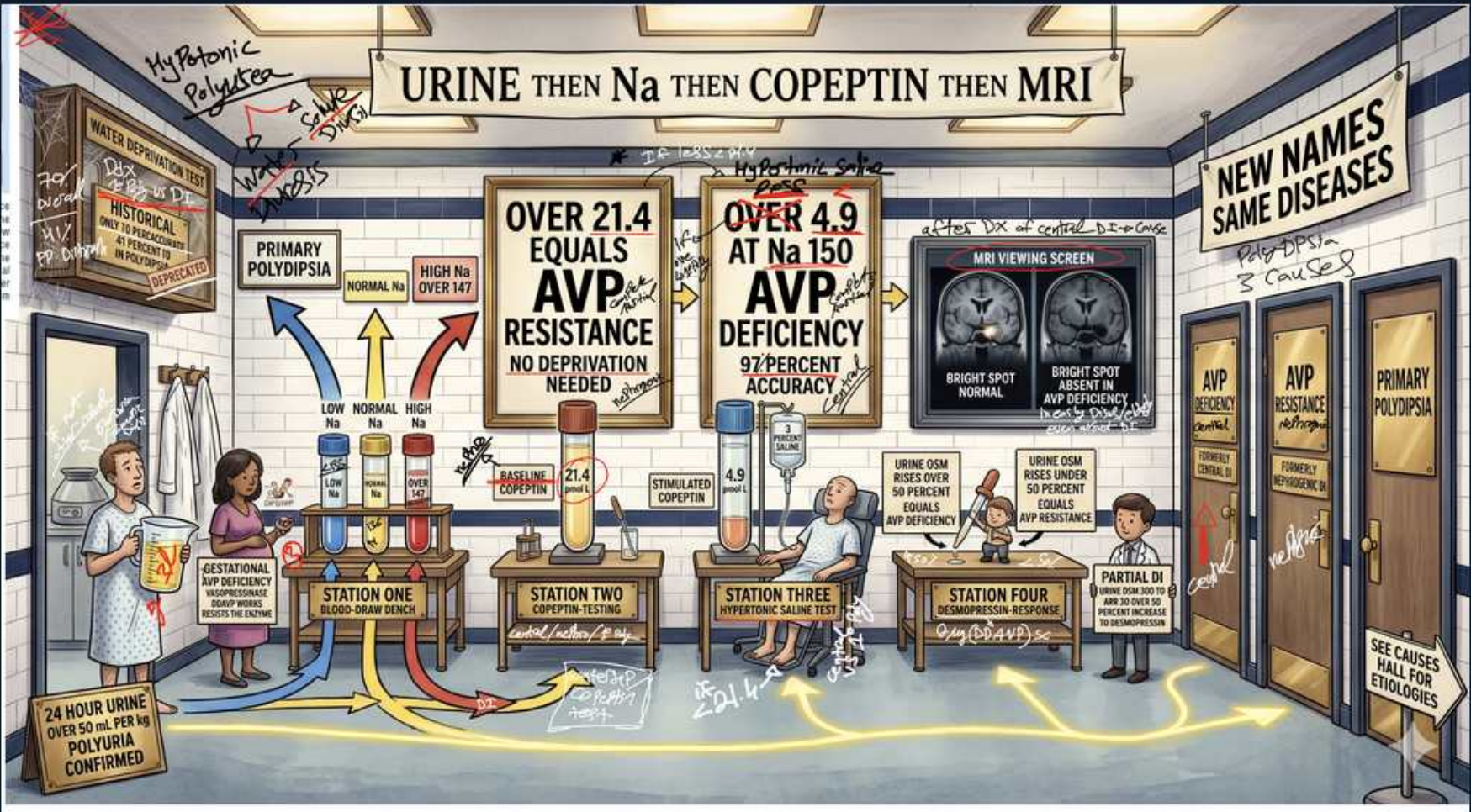


DI Workup — Copeptin Cutoffs & MRI Bright Spot





## 1. Urine → Na → Copeptin → MRI

### WHAT YOU SEE

Banner 'URINE THEN Na THEN COPEPTIN THEN MRI'.

### WHAT IT ENCODES

The modern polyuria algorithm: confirm true polyuria on 24h urine, check serum sodium, then use copeptin (a stable surrogate for AVP), and finally MRI to localize central causes. Copeptin has largely replaced the classic water-deprivation test.

### BOARD TRAP

Sequence matters — jumping straight to MRI without copeptin/Na can mislabel primary polydipsia as central DI.

## 2. 24h urine confirms polyuria

### WHAT YOU SEE

Bottom-left '24 hour urine, polyuria confirmed'.

### WHAT IT ENCODES

Document urine output >3 L/day before any provocative testing. Distinguishes genuine polyuria from frequency and anchors the whole pathway.

### BOARD TRAP

Frequency without high volume is not polyuria — skipping the collection wastes copeptin/MRI on the wrong patient.

## 3. Baseline copeptin (Station One)

### WHAT YOU SEE

Station One: baseline blood-draw tube, '21.4'.

### WHAT IT ENCODES

Baseline (unstimulated) copeptin is drawn first. A very high baseline copeptin (>21.4 pmol/L) without prior fluid restriction indicates nephrogenic DI — the kidney resists abundant AVP. No water deprivation needed at that level.

### BOARD TRAP

A sky-high baseline copeptin means nephrogenic DI (AVP resistance), NOT deficiency — don't proceed to deprivation when the level is already maximal.

## 4. Copeptin >21.4 = AVP resistance (nephrogenic)

### WHAT YOU SEE

Orange tube 'OVER 21.4 EQUALS AVP RESISTANCE, no deprivation needed'.

### WHAT IT ENCODES

Baseline copeptin >21.4 pmol/L diagnoses nephrogenic DI (AVP resistance) outright. The high endogenous AVP that the failing kidney can't respond to drives copeptin up. Spares the patient deprivation testing.

### BOARD TRAP

AVP resistance (nephrogenic) shows HIGH copeptin; AVP deficiency (central) shows LOW — flip these and you'll give DDAVP to a resistant kidney that won't respond.

## 5. Copeptin >4.9 at Na 150 = AVP deficiency (central)

### WHAT YOU SEE

Orange tube 'OVER 4.9 AT Na 150 = AVP DEFICIENCY, 97% accuracy'.

### WHAT IT ENCODES

Hypertonic saline–stimulated copeptin: at serum Na ~150 mmol/L, a copeptin BELOW ~4.9 pmol/L indicates central DI (AVP deficiency), while a value above it indicates primary polydipsia. ~97% diagnostic accuracy — superior to water deprivation.

### BOARD TRAP

The 4.9 cutoff is read AT the target sodium (~150) under stimulation — interpreting an unstimulated value against it misclassifies primary polydipsia as central DI.

## 6. Stimulated saline test (Station Three)

### WHAT YOU SEE

Station Three 'stimulated saline test', patient in chair.

### WHAT IT ENCODES

Hypertonic (3%) saline raises serum sodium to a hyperosmolar target (~150) to maximally stimulate AVP/copeptin release. Requires close Na monitoring. Reads against the 4.9 cutoff to separate central DI from primary polydipsia.

### BOARD TRAP

Hypertonic saline stimulation needs frequent sodium checks — overshooting risks dangerous hyponatremia, so it is done under supervision.

## 7. MRI viewing screen — pituitary bright spot

### WHAT YOU SEE

MRI screen: bright spot normal vs absent.

### WHAT IT ENCODES

T1 posterior pituitary 'bright spot' reflects stored AVP. Present (normal) bright spot vs absent bright spot in central AVP deficiency. MRI localizes tumor/infiltration/stalk lesions after copeptin confirms central DI.

### BOARD TRAP

An absent bright spot supports central DI but is not 100% specific (can fade with age) — confirm with copeptin, don't diagnose on MRI alone.

## 8. Urine osm > 50% rise after DDAVP = central

### WHAT YOU SEE

Station Four 'urine osm over 50% rise after DDAVP = AVP deficiency'.

### WHAT IT ENCODES

After confirming inability to concentrate, give DDAVP: urine osmolality rising >50% indicates central DI (the kidney responds to replaced AVP). Minimal rise indicates nephrogenic DI.

### BOARD TRAP

>50% concentration after DDAVP = central (responsive); <50% = nephrogenic (resistant) — the magnitude of response, not just any change, is the discriminator.



## 9. Partial DI — urine osm 10–50% rise

### WHAT YOU SEE

Partial DI note '10–50%... incomplete'.

### WHAT IT ENCODES

Partial central or partial nephrogenic DI gives an intermediate (~10–50%) urine osm response to DDAVP, overlapping with primary polydipsia. Copeptin stimulation resolves these ambiguous partial cases best.

### BOARD TRAP

Intermediate DDAVP responses are the gray zone where water deprivation fails — copeptin is specifically more accurate for partial DI.

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## 10. New names, same diseases

### WHAT YOU SEE

Sign 'NEW NAMES SAME DISEASES': AVP deficiency / AVP resistance / primary polydipsia.

### WHAT IT ENCODES

Updated nomenclature: 'AVP deficiency' = central DI, 'AVP resistance' = nephrogenic DI; primary polydipsia unchanged. Names reflect mechanism and reduce dangerous confusion with diabetes mellitus.

### BOARD TRAP

AVP deficiency = central, AVP resistance = nephrogenic — mixing the new terms up reverses the treatment (DDAVP helps deficiency, not resistance).

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## 11. Hyponatremic polydipsia clue

### WHAT YOU SEE

Handwritten 'hypotonic polydipsia' near top-left.

### WHAT IT ENCODES

Primary polydipsia typically presents with low-normal/low serum sodium from water overload and suppressed AVP, whereas DI trends toward high-normal sodium. Baseline Na frames the copeptin interpretation.

### BOARD TRAP

Low serum Na before testing favors primary polydipsia; high-normal Na favors true DI — the starting sodium steers the whole workup.

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## 12. Copeptin = stable AVP surrogate (Station Two)

### WHAT YOU SEE

Station Two 'copeptin testing' tubes.

### WHAT IT ENCODES

Copeptin is the C-terminal fragment of pro-vasopressin, co-secreted 1:1 with AVP but far more stable in plasma, making it measurable where AVP assays fail. This stability is why copeptin replaced direct AVP measurement.

### BOARD TRAP

AVP itself is unstable and hard to assay — relying on a direct AVP level instead of copeptin gives unreliable results.

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